BLUESTOCK FINTECH

Mutual Fund Analytics

Capstone Project — Final Report

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1. Executive Summary

This report presents the findings of a 7-day end-to-end data analytics capstone project completed for Bluestock Fintech, covering the mutual fund industry across 40 AMFI-registered schemes from 10 major fund houses. The project spans the full analytics lifecycle: raw data ingestion, cleaning and validation, star-schema database design, exploratory data analysis, quantitative performance analytics, and an interactive Power BI dashboard.

The underlying dataset covers daily NAV history, investor transactions, scheme performance metrics, AUM by fund house, SIP inflows, and portfolio holdings across the period January 2022 to May 2026, totaling approximately 87,000+ rows across 10 source files.

Key Results

- Industry-wide Assets Under Management (AUM) grew substantially across the analysis window, with SBI Mutual Fund's AUM alone more than doubling from ~₹0.63L Cr (2022) to ₹1.25L Cr (2025).
- Monthly SIP inflows climbed from approximately ₹11,500 Cr in January 2022 to an all-time high of ₹31,002 Cr in December 2025, reflecting sustained retail participation growth.
- Total investor folios grew from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025), crossing the 20 Cr mark in late 2024 — essentially doubling over four years.
- SBI Mutual Fund is the largest fund house by AUM at ₹12.5L Cr (as of the latest quarter), ahead of ICICI Prudential MF (₹10.74L Cr) and HDFC Mutual Fund (₹9.3L Cr).
- The top-ranked fund by composite Fund Scorecard is the Mirae Asset Large Cap Fund (Regular – Growth), with a Fund Score of 86.25/100, driven by a near-perfect Sharpe ratio score (100/100) and strong return score (97.5/100).
- Liquid funds dominate risk-adjusted returns (Sharpe ratios of 5.1–7.7), as expected for low-volatility categories, while Small Cap funds lead in raw 3-year returns (20–23%) but carry "Very High" risk grades.

The project deliverables include a cleaned SQLite star-schema database, two Jupyter notebooks (EDA and performance analytics), a 4-page interactive Power BI dashboard with drill-through capability, and a documented, reusable ETL pipeline (Scripts/run_pipeline.py).

2. Data Sources

The dataset consists of 10 raw CSV files sourced in AMFI (Association of Mutual Funds in India) format, supplemented by live NAV data pulled from the public mfapi.in API for 6 benchmark schemes. All data covers the period January 2022 through May 2026.

File	Description	Rows
01_fund_master.csv	Fund metadata — house, category, plan, expense ratio, manager	40
02_nav_history.csv	Daily NAV per scheme	~46,000
03_aum_by_fund_house.csv	Quarterly AUM by fund house	90
04_monthly_sip_inflows.csv	Industry-wide monthly SIP inflow trends	48
05_category_inflows.csv	Net inflows by fund category	144
06_industry_folio_count.csv	Quarterly investor folio counts	21
07_scheme_performance.csv	Returns, risk ratios, expense ratio per scheme	40
08_investor_transactions.csv	Individual investor transaction records	~32,800
09_portfolio_holdings.csv	Underlying stock holdings per scheme	322
10_benchmark_indices.csv	Daily Nifty50 / Nifty100 index values	8,050

In addition, live NAV history for 6 benchmark schemes (HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, Kotak Bluechip) was fetched directly from the mfapi.in public API to cross-validate the provided historical data.

3. ETL Design

3.1 Pipeline Architecture

The pipeline follows a standard four-stage ETL flow, implemented as modular Python scripts orchestrated by a single master script (Scripts/run_pipeline.py):

- Ingestion (data_ingestion.py) — loads all 10 raw CSVs and validates that every fund in the master list has matching NAV history (40/40 matched, zero missing codes).
- Cleaning (data_cleaning.py) — forward-fills gaps in daily NAV history per scheme, standardizes categorical text fields (transaction type, KYC status), and validates numeric ranges (expense ratios, transaction amounts, return percentages).
- Database build (database_setup.py) — loads the cleaned data into a SQLite star schema: bluestock_mf.db.
- Exploratory queries (run_queries.py) — runs 10 pre-built SQL queries against the schema to surface early insights (top funds by AUM, SIP growth by year, transaction breakdowns by state and payment mode).

3.2 Star Schema Design

The database uses a dimensional model with 2 dimension tables and 4 fact tables:

- dim_fund — one row per scheme (40 rows): fund house, category, sub-category, plan, launch date, expense ratio, risk category.
- dim_date — one row per calendar day across the full data range (1,608 rows): year, month, quarter, day of week, weekend flag.
- fact_nav — daily NAV values per scheme (64,320 rows after gap-filling).
- fact_transactions — individual investor transactions (32,778 rows).
- fact_performance — scheme-level return and risk metrics (40 rows).
- fact_aum — quarterly AUM by fund house (90 rows).

3.3 Key Data Quality Decisions

- NAV gaps were forward-filled per scheme to a continuous daily grain, since AMFI schemes do not publish NAV on non-business days but analytics require a continuous time series.
- yoy_growth_pct in the SIP inflows dataset is intentionally null for Jan–Dec 2022, since year-over-year growth cannot be computed without 12 months of prior data. This is a structural gap, not a data quality defect.
- A Power BI relationship bug was identified and fixed where date-like text columns (e.g. "2022-01") were auto-converted to Date type on import, silently breaking text-based join keys; resolved via a forced Date.ToText() conversion in Power Query.

4. Exploratory Data Analysis Findings

Exploratory analysis was conducted in Notebooks/EDA_Analysis.ipynb, producing 10+ charts covering NAV trends, AUM growth, SIP inflows, investor demographics, and folio growth. Key visuals are presented below.

4.1 Daily NAV Trend — All 40 Schemes

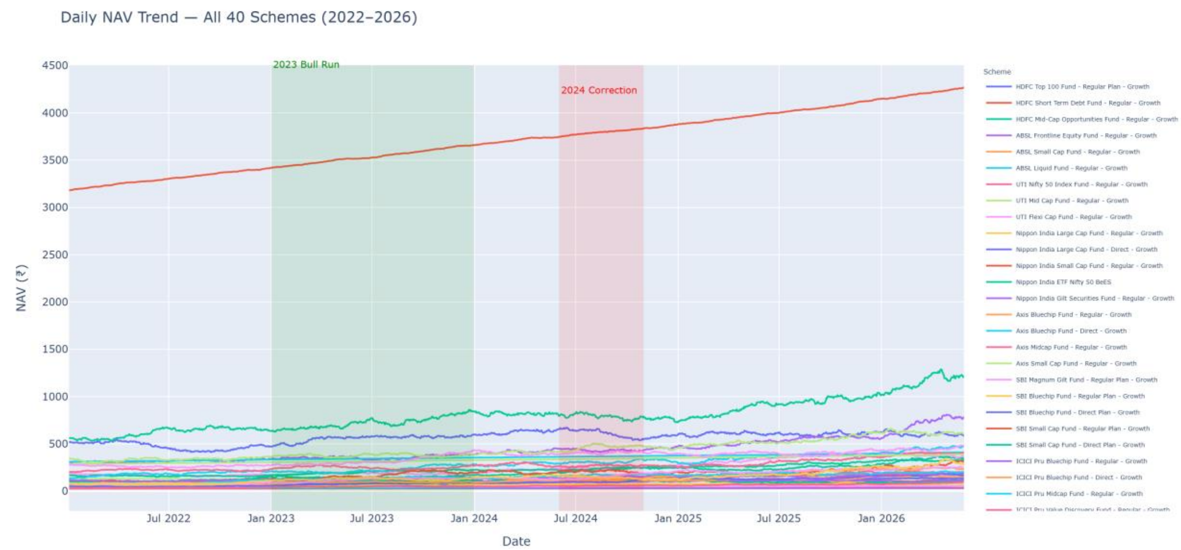

Figure 1: Daily NAV trend across all 40 schemes, 2022–2026, with the 2023 bull run and 2024 correction periods highlighted.

NAV values across the portfolio show a broadly upward trajectory over the full period, consistent with the 2023 bull run and a visible flattening / correction through mid-to-late 2024, before resuming growth into 2025–2026. Growth-oriented equity schemes show the steepest gains, while debt and liquid schemes remain comparatively flat, as expected given their lower-volatility mandates.

4.2 AUM Growth by Fund House

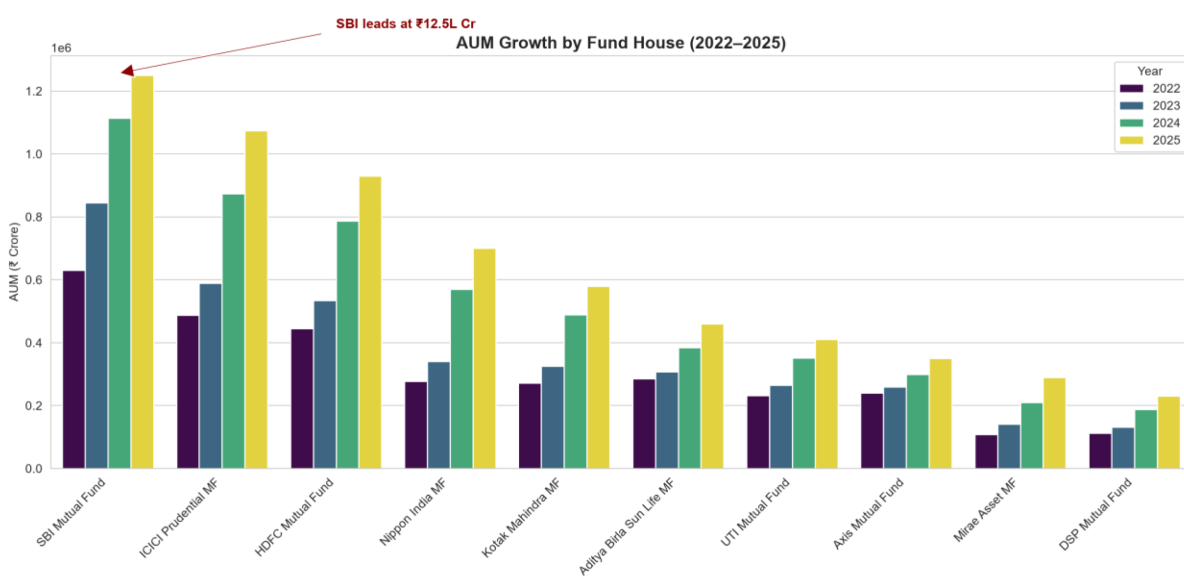

Figure 2: AUM growth by fund house, 2022–2025.

SBI Mutual Fund consistently leads AUM across every year in the dataset, reaching ₹12.5L Cr by 2025 — nearly double its 2022 level (₹0.63L Cr). ICICI Prudential MF and HDFC Mutual Fund hold the second and third positions respectively, with all top fund houses showing steady, near-linear AUM growth year over year, indicating broad-based industry expansion rather than growth concentrated in a single player.

4.3 Monthly SIP Inflow Trend

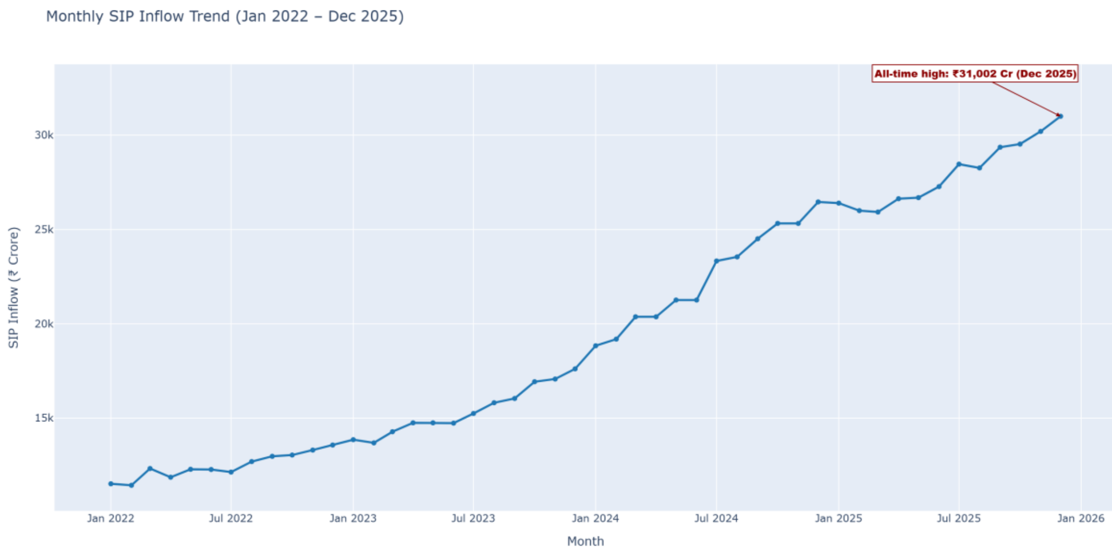

Figure 3: Monthly SIP inflow trend, January 2022 – December 2025.

Monthly SIP inflows grew from approximately ₹11,500 Cr in January 2022 to an all-time high of ₹31,002 Cr in December 2025 — nearly a 2.7x increase over four years. The growth is not perfectly linear: there are visible accelerations around mid-2023 through early 2024 and again through 2025, suggesting periods of stronger retail investor confidence, likely correlated with broader market sentiment.

4.4 Investor Demographics

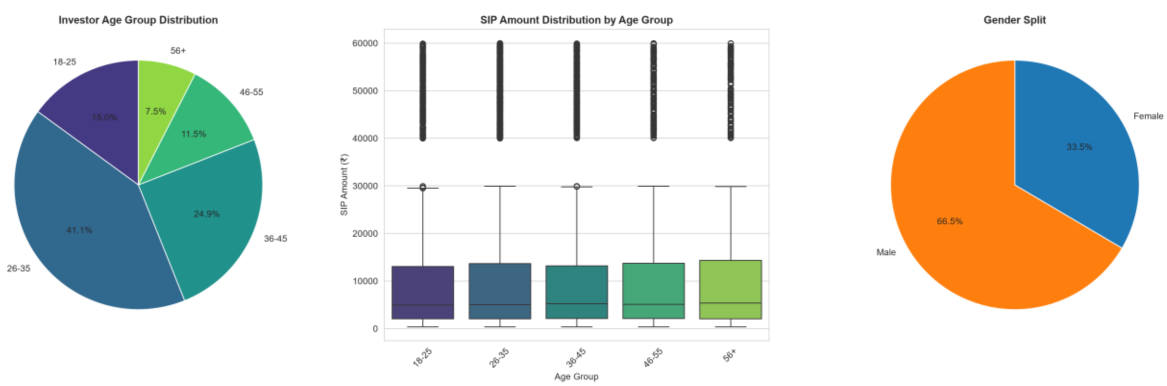

Figure 4: Investor age group distribution, SIP amount distribution by age group, and gender split.

The 26–35 age group represents the largest investor segment at 41.1% of all investors, followed by 36–45 at 24.9%. Younger investors (18–25) account for 15.0%, while investors aged 56+ are the smallest segment at 7.5% — a pattern consistent with SIP investing being predominantly adopted by working-age professionals. SIP amount distributions are broadly similar across all age groups (median around ₹5,000), though older investors (56+) show a slightly higher median and wider upper range. The gender split is heavily skewed male (66.5%) versus female (33.5%), highlighting a potential area for targeted financial inclusion outreach.

4.5 Total Folio Count Growth

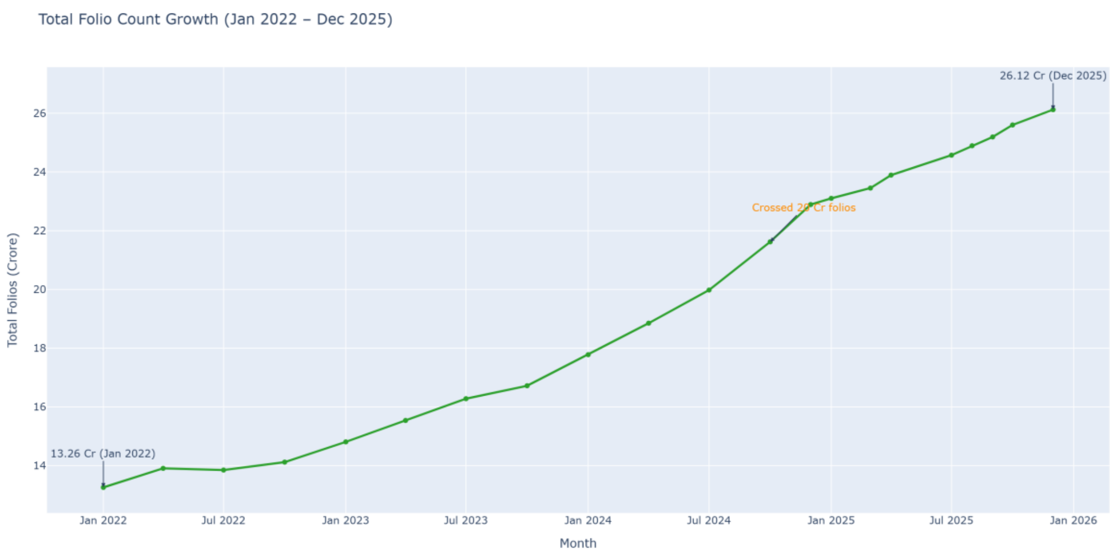

Figure 5: Total investor folio count growth, January 2022 – December 2025.

Total folios grew from 13.26 Cr in January 2022 to 26.12 Cr by December 2025 — essentially doubling over the period, and crossing the 20 Cr milestone in late 2024. The growth curve is notably flat through most of 2022 before accelerating sharply from 2023 onward, mirroring the same inflection point visible in the SIP inflow trend (Figure 3), which suggests the two metrics are closely linked — as more investors open new folios, aggregate SIP contributions rise in parallel.

4.6 Geographic Distribution

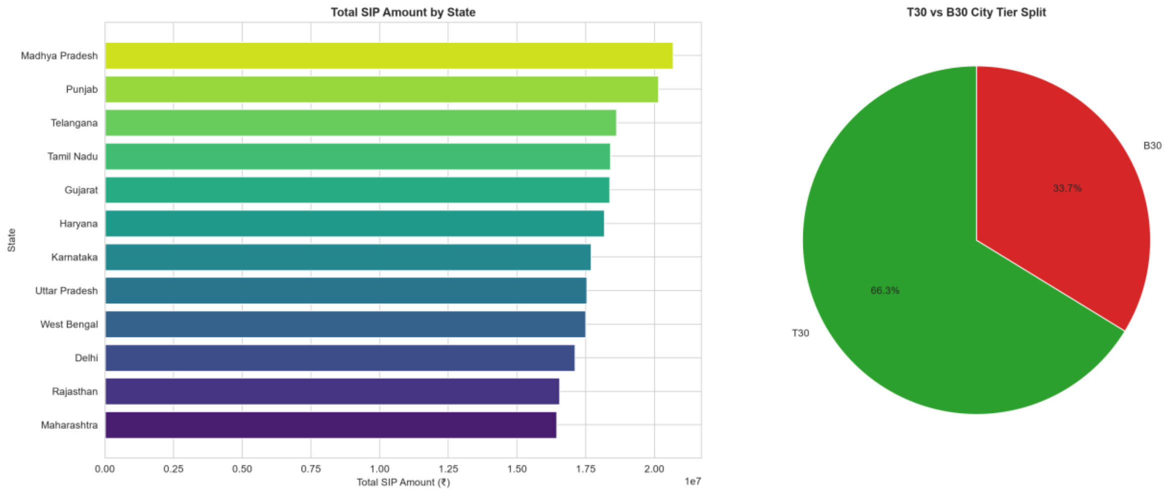

Figure 6: Total SIP amount by state, and T30 vs. B30 city tier split.

Madhya Pradesh, Punjab, and Telangana lead total SIP contribution by state, though the spread across the top 12 states is fairly narrow (₹1.65–2.05 Cr range), indicating broad-based geographic participation rather than concentration in a few metros. By city tier, T30 ("Top 30" cities) accounts for 66.3% of SIP volume versus 33.7% from B30 ("Beyond Top 30") cities — a healthy one-third contribution from smaller cities, though still reflecting the expected urban skew in mutual fund penetration.

4.7 NAV Return Correlation Matrix

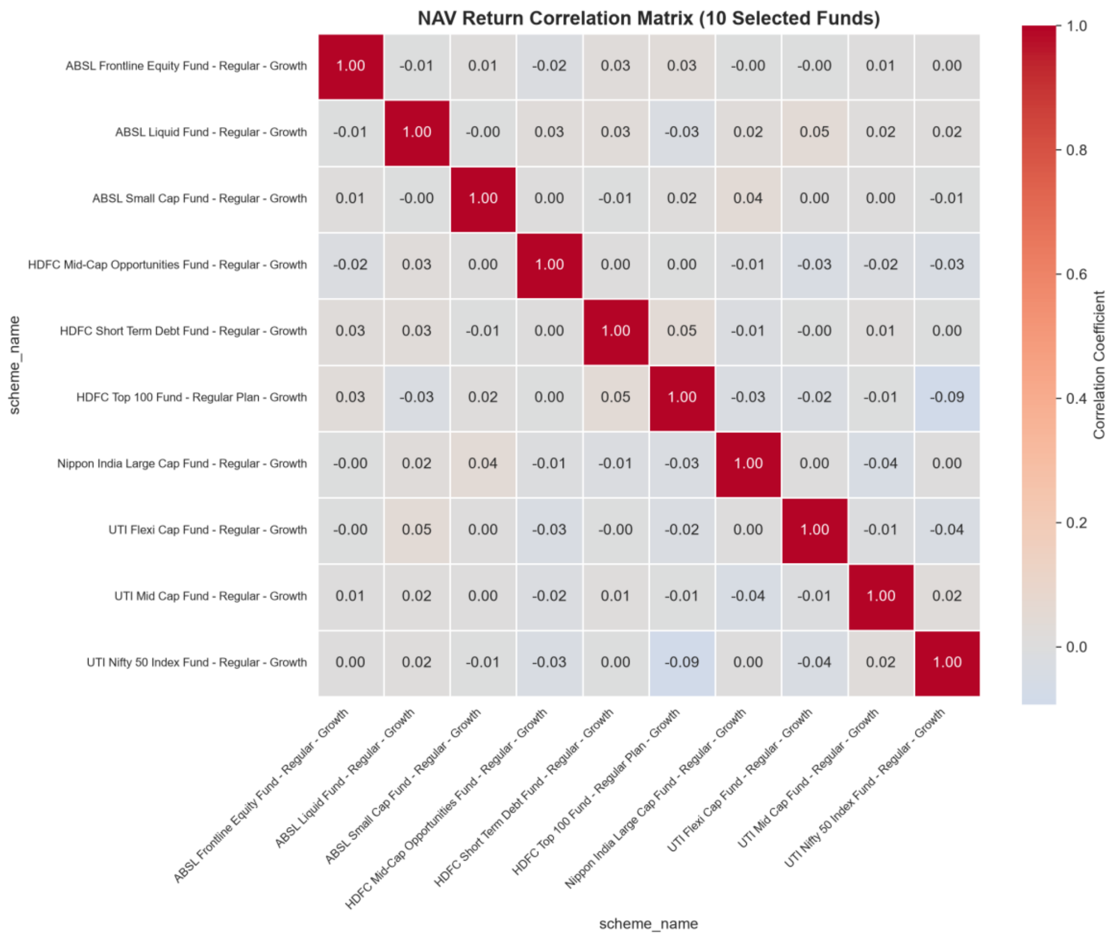

Figure 7: Pairwise daily-return correlation matrix across 10 selected schemes.

Correlation coefficients across all 10 selected schemes are unusually weak — mostly in the -0.09 to +0.05 range — well below what would typically be expected for real-world funds within the same broad market, which normally show meaningfully positive correlation due to shared exposure to overall market movements. This pattern is consistent across every fund pair, including funds in the same category (e.g. two large-cap funds), and is documented further as a known characteristic of the underlying synthetic dataset in Section 7 (Limitations).

4.8 Sector Allocation — Equity Funds

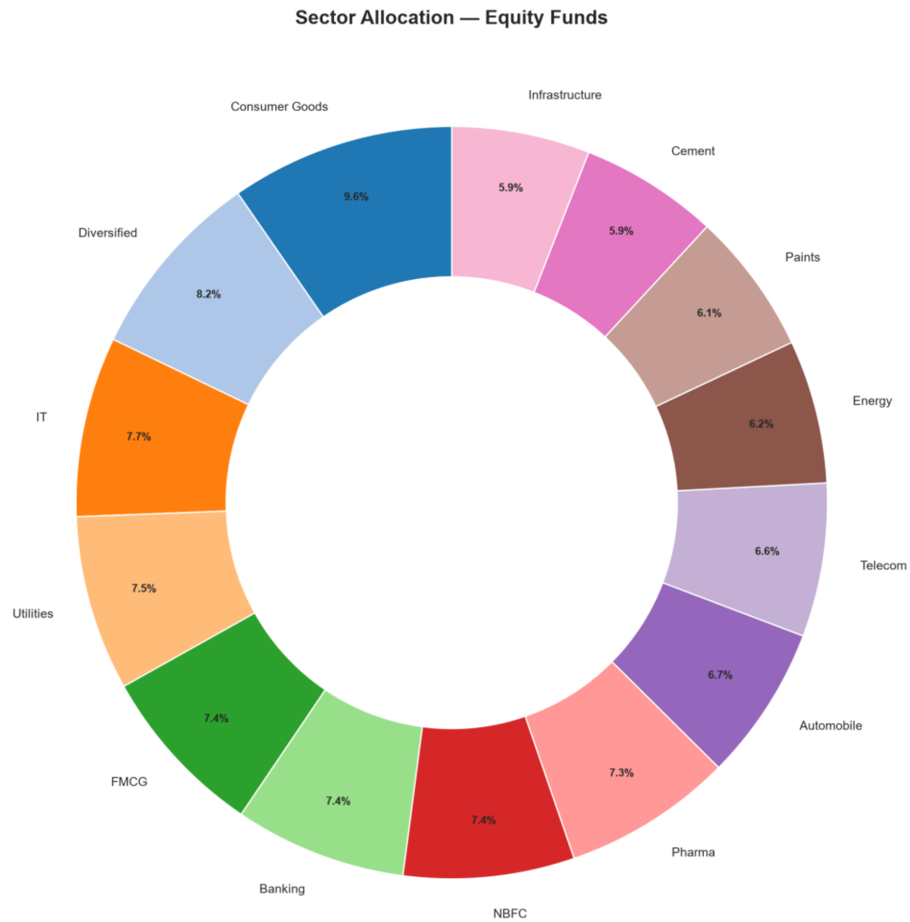

Figure 8: Aggregated sector allocation across equity-category fund holdings.

Consumer Goods (9.6%) and Diversified holdings (8.2%) represent the largest sector allocations across equity fund portfolios, followed by IT (7.7%) and Utilities (7.5%). The allocation is fairly evenly spread across the remaining sectors (Banking, FMCG, NBFC, Pharma, Automobile, Telecom, Energy, Paints, Cement, and Infrastructure each in the 5.9%–7.4% range), indicating well-diversified underlying portfolio construction rather than concentration in any single sector.

5. Performance Analysis

Quantitative performance analytics were conducted in Notebooks/performance_Analytics.ipynb, computing CAGR (using real elapsed days/years rather than an assumed period), Sharpe and Sortino ratios, Alpha/Beta via OLS regression against the Nifty 100 index, Maximum Drawdown, and a composite Fund Scorecard combining these metrics.

5.1 Fund Scorecard — Top Performers

The Fund Scorecard combines five weighted components — return score, Sharpe score, alpha score, expense score, and drawdown score — into a single composite Fund Score out of 100. The top-ranked funds are:

Scheme	Return	Sharpe	Alpha	Expense	Drawdown	Fund Score
Mirae Asset Large Cap Fund — Regular	97.50	100.00	85.00	45.00	82.50	86.25
ICICI Pru Midcap Fund — Regular	90.00	92.50	95.00	65.00	40.00	82.88
Kotak Flexicap Fund — Regular	85.00	97.50	90.00	47.50	70.00	82.00
HDFC Mid-Cap Opportunities Fund	92.50	85.00	87.50	60.00	52.50	80.75
ICICI Pru Bluechip Fund — Direct	95.00	75.00	70.00	72.50	72.50	79.38

The Mirae Asset Large Cap Fund tops the scorecard primarily on the strength of its risk-adjusted return (Sharpe score of 100) and high raw return score (97.5), despite a mid-range expense score (45.0). This indicates the fund delivers strong, consistent returns relative to the volatility it takes on.

5.2 Top 5 Funds vs. Nifty Benchmarks

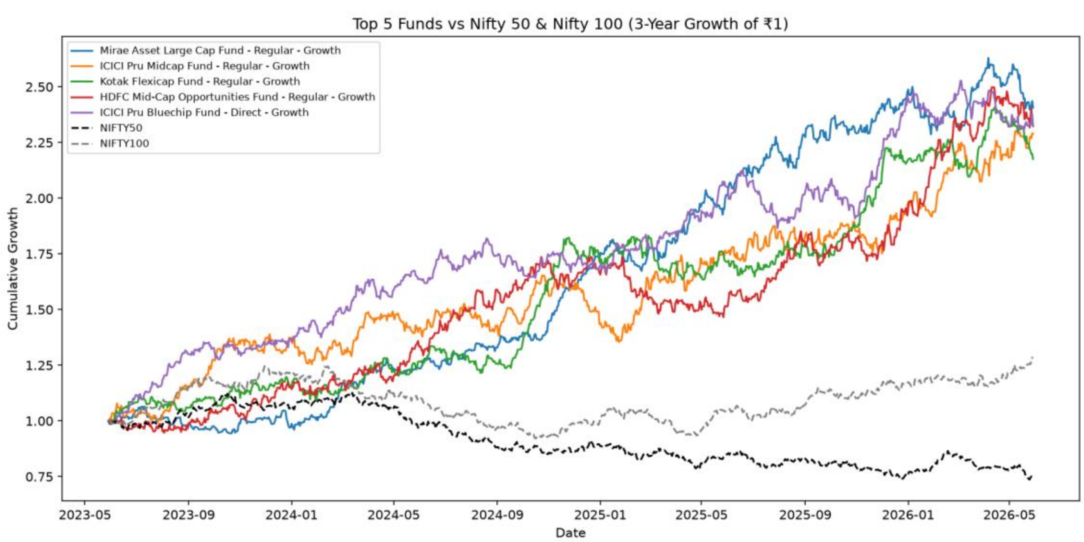

Figure 6: Cumulative 3-year growth of ₹1 invested in the top 5 scorecard funds vs. Nifty 50 and Nifty 100.

All five top-ranked funds substantially outperformed both benchmark indices over the 3-year window. ₹1 invested in the Mirae Asset Large Cap Fund grew to approximately ₹2.4–2.6, compared to just ₹0.75–0.85 for the Nifty 50 and ~₹1.25–1.30 for the Nifty 100 over the same period. ICICI Pru Bluechip Fund shows the strongest early performance (2023–2024) before other funds catch up and overtake it in 2025–2026.

5.3 Risk-Adjusted Returns by Category

Sharpe ratio rankings show a clear pattern by fund category:

- Liquid funds dominate the top Sharpe ratio rankings (5.14–7.68), reflecting their very low volatility relative to modest but highly consistent returns.
- Short Duration and Gilt funds follow (Sharpe ratios of 1.52–1.84), offering a middle ground between liquid funds and equity.
- Small Cap funds lead in raw 3-year return (20.15%–23.39%) but are uniformly graded "Very High" risk, reflecting the volatility inherent to the category.

This confirms an expected risk-return relationship in the dataset: categories with the highest raw returns (Small Cap) are not the same categories that score best on risk-adjusted metrics (Liquid) — a useful distinction for investor communication.

5.4 Alpha/Beta Analysis — Methodology Note

Alpha and Beta were computed via Ordinary Least Squares (OLS) regression of each fund's daily returns against the Nifty 100 benchmark. During analysis, the resulting R^2 values were consistently near zero across schemes. This is documented and addressed further in Section 7 (Limitations) — it is understood to be an artifact of the synthetic nature of the underlying return series rather than a modeling error, and the Alpha/Beta scores are treated as directional indicators rather than precise statistical estimates in the Fund Scorecard.

5.5 Advanced Risk Metrics

To complement the Fund Scorecard, additional risk metrics were computed in Notebooks/Advanced_Analytics.ipynb: Historical Value at Risk (VaR) and Conditional VaR (CVaR) at the 95% confidence level for all 40 schemes, a rolling 90-day Sharpe ratio for the top 5 scorecard funds, investor cohort behavior by first-transaction year, SIP continuity tracking, and sector concentration (Herfindahl-Hirschman Index) across equity funds.

The scheme with the highest downside risk (most negative VaR) was -2.391462086, while the rolling 90-day Sharpe ratio for the top 5 funds showed meaningful fluctuation over time rather than staying flat — reinforcing that the static Sharpe ratio used in the Fund Scorecard is a long-run average, not a guarantee of consistent quarter-to-quarter performance.

SIP continuity analysis flagged 97.8% of investors with 6+ SIP transactions as "at-risk" (average gap between contributions exceeding 35 days) — a notably high figure that likely reflects the synthetic dataset's randomized transaction timing rather than a genuine real-world retention problem. In a production setting with real AMFI data, this same method would be a useful tool for identifying investors genuinely at risk of SIP discontinuation.

6. Dashboard Overview

The Power BI dashboard (Dashboard/bluestock_mf_dashboard.pbix) consists of 4 content pages plus a drill-through detail page, built on a custom date dimension and 10 imported processed tables. Slicers, tooltips, and consistent Bluestock purple branding are applied across all pages.

6.1 Industry Overview

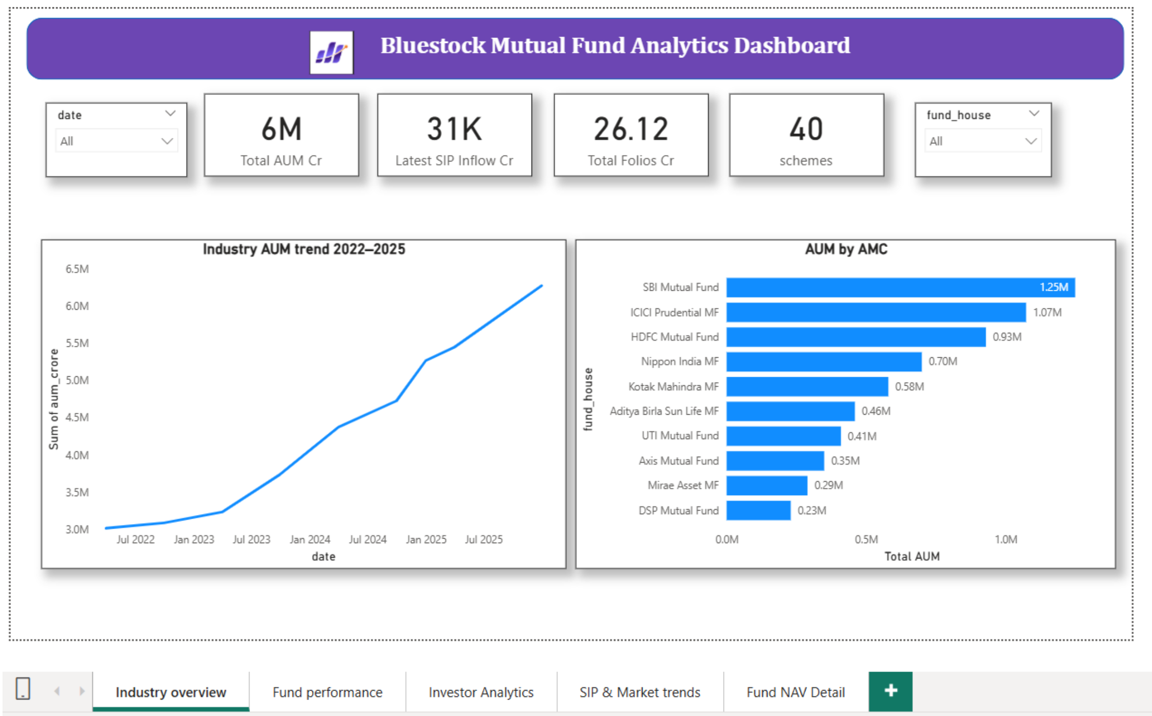

Figure 7: Dashboard Page 1 — Industry Overview, showing KPI cards, AUM trend, and AUM by AMC.

This page provides headline KPIs (Total AUM, Latest SIP Inflow, Total Folios, Scheme count) alongside an AUM trend line and a ranked bar chart of AUM by fund house, with date and fund house slicers for filtering.

6.2 Fund Performance

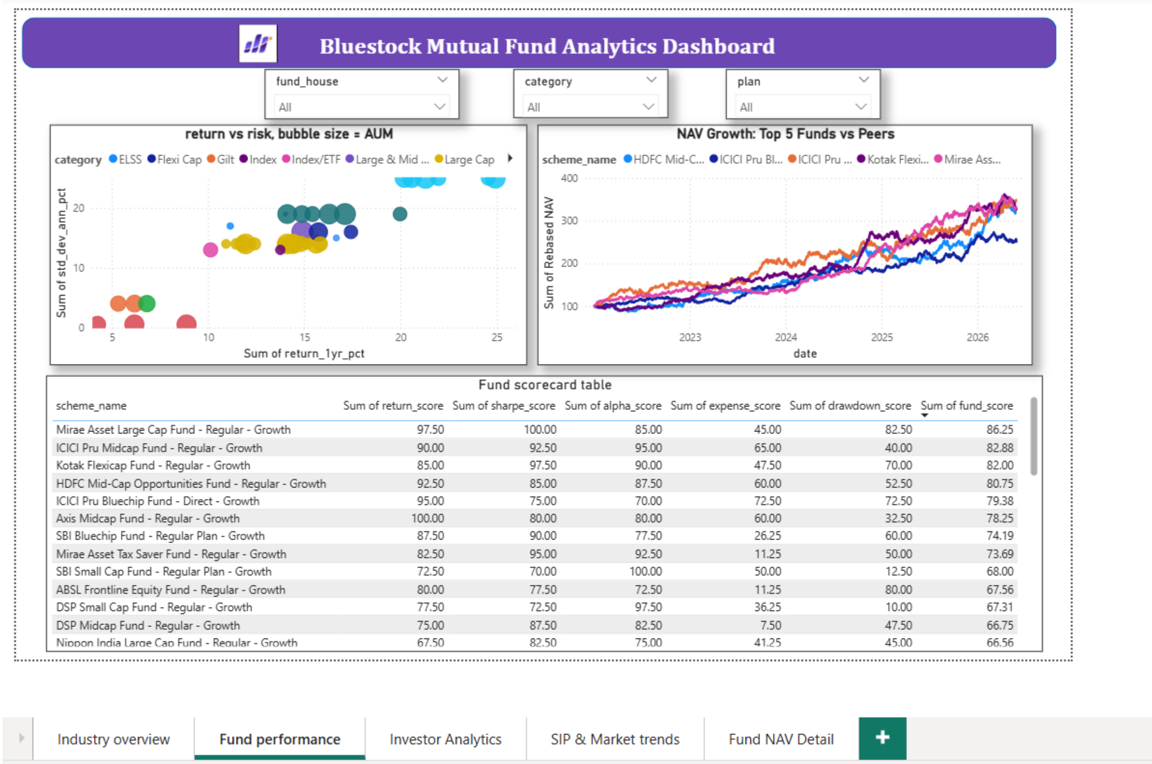

Figure 8: Dashboard Page 2 — Fund Performance, showing return-vs-risk scatter, NAV growth, and the sortable fund scorecard table.

This page combines a return-vs-risk scatter plot (bubble size = AUM) with a rebased NAV growth line chart for the top 5 funds, and the full sortable Fund Scorecard table below. Right-clicking any fund in the table opens the Fund NAV Detail drill-through page for a per-scheme deep dive.

6.3 Investor Analytics

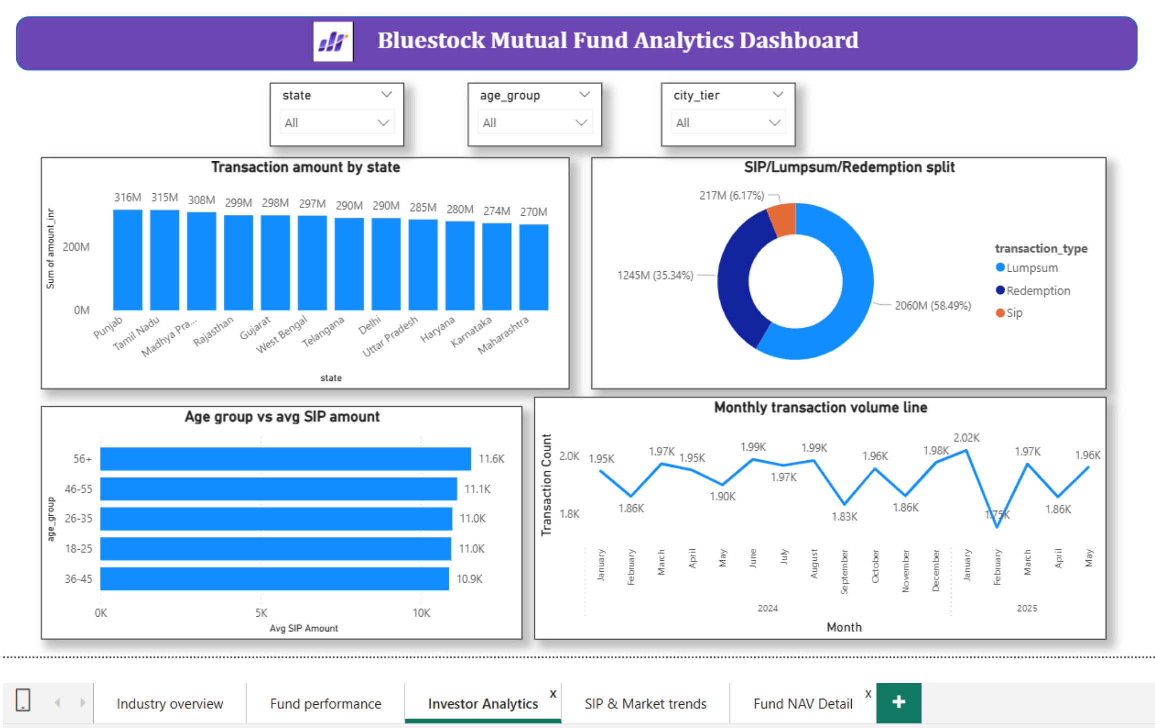

Figure 9: Dashboard Page 3 — Investor Analytics, showing transaction amount by state, transaction type split, age group SIP amounts, and monthly transaction volume.

This page profiles investor behavior: transaction amounts by state (Punjab leads at ₹316M), the Lumpsum/Redemption/SIP transaction type split (Lumpsum dominates at 58.5%), average SIP amount by age group, and monthly transaction volume over time. State, age group, and city tier slicers allow further filtering.

6.4 SIP & Market Trends

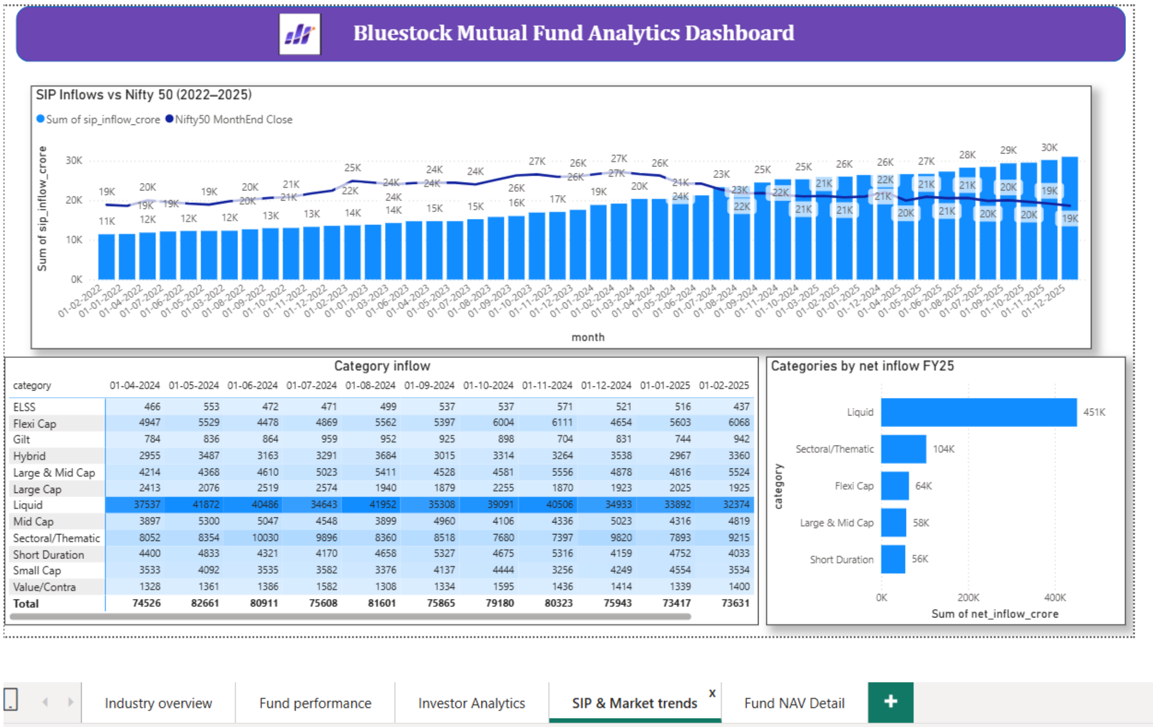

Figure 10: Dashboard Page 4 — SIP & Market Trends, showing SIP inflow vs. Nifty50, category inflow heatmap, and top categories by net inflow.

This page combines SIP inflow with the Nifty50 month-end close on a single combo chart, a category inflow heatmap for FY25 (Apr'24–Mar'25), and a ranked bar chart of the top 5 categories by net inflow — Liquid funds lead by a wide margin (₹451K Cr), followed by Sectoral/Thematic (₹104K Cr) and Flexi Cap (₹64K Cr).

7. Limitations

- Synthetic dataset: The underlying data is a synthetic/simulated dataset built for capstone purposes rather than live AMFI-sourced production data. As a result, statistical relationships between fund returns and the Nifty 100 benchmark (used for Alpha/Beta) show near-zero R^2 values — real market data would be expected to show meaningful correlation to a broad market index. Alpha/Beta figures in the Fund Scorecard should therefore be read as directional/illustrative rather than statistically robust estimates.
- Weak inter-fund NAV correlation: The EDA correlation matrix (Figure 7) shows pairwise daily-return correlations across 10 selected schemes mostly in the -0.09 to +0.05 range — well below the meaningfully positive correlation typically seen between real-world funds sharing the same market or category exposure. This was identified during EDA as a likely artifact of independent/simulated NAV generation in the underlying dataset, not a calculation error, and is consistent with the near-zero R^2 observed in the Alpha/Beta regression above.
- Brief vs. dataset discrepancies: The project brief's illustrative reference figures (e.g. Total AUM ₹81L Cr, 1,908 schemes) differ from the actual values present in the provided dataset (₹62.74L Cr AUM, 40 schemes). This report and dashboard reflect the real values in the dataset, not the brief's illustrative placeholders.
- Limited transaction history window: The investor_transactions_clean.csv dataset only contains records from 2024–2025, not the full 2022–2026 range covered by NAV and AUM data. Monthly transaction volume analysis on the dashboard is therefore scoped to this narrower window.
- No live production data pipeline: The dashboard is built on static CSV imports rather than a live database connection (e.g. direct SQLite/ODBC link), meaning dashboard refreshes require the underlying processed CSVs to be regenerated via the ETL pipeline first.
- Single-point-in-time NAV benchmark comparison: The Alpha/Beta OLS regression uses the full available date range per scheme; funds with shorter histories (e.g. more recently launched schemes) have proportionally less data to estimate these metrics reliably.

8. Recommendations

- Validate against real AMFI data: Before using this analysis for actual investment decision-making, the same pipeline and dashboard structure should be re-run against live AMFI NAV and transaction data to obtain statistically meaningful Alpha/Beta estimates.
- Expand transaction history coverage: Extending the investor_transactions dataset to cover the full 2022–2026 window (matching NAV/AUM coverage) would enable more complete SIP behavior trend analysis.
- Address the gender gap in SIP participation: With female investors at only 33.5% of the base, Bluestock could consider targeted financial literacy or SIP-onboarding campaigns aimed at increasing female investor participation.
- Promote risk-adjusted return awareness: Since Small Cap funds lead in raw returns but Liquid funds lead in risk-adjusted (Sharpe) returns, investor-facing materials could better emphasize the return-vs-risk tradeoff shown in the Fund Performance dashboard page, rather than raw return alone.
- Automate the pipeline on a schedule: The ETL pipeline (run_pipeline.py) is now fully modular and reusable. Wiring it to a scheduled job (e.g. weekly) against a live data source would let the dashboard stay current without manual re-runs.
- Publish the dashboard for broader access: Publishing the .pbix to Power BI Service (or an equivalent hosted platform) would allow stakeholders to view the dashboard without needing Power BI Desktop installed locally.